

## Graphing Lines, Intersections, &amp; Inequalities — Concept Focus

## Mini-Review (read first)

**Line:**  $y = mx + b$  where  $m$  is slope,  $b$  is  $y$ -intercept. Every point  $(x, y)$  on the graph satisfies the equation.

**System of two lines:** Any intersection point solves *both* equations simultaneously.

• One intersection  $\Rightarrow$  one solution. • Parallel distinct  $\Rightarrow$  no solution. • Same line  $\Rightarrow$  infinitely many.

**Half-plane:** An inequality like  $y > mx + b$  means points *above* the line  $y = mx + b$  (line dashed).

$y \geq mx + b$  includes the boundary (solid line). Test a point to decide which side to shade.

**Modeling:** Translate words  $\rightarrow$  equations/inequalities; the solution set is where all constraints hold at once.

1. (**Concept**) Explain in one or two sentences why the intersection point of two non-parallel lines is the solution to the system formed by their equations.

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2. (**True/False + why**) T/F: “If a point lies on the line  $y = 2x - 3$ , then it is a solution to  $2x - y = 3$ .” Justify.

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3. (**Find by logic**) The line  $\ell_1$  passes through  $(0, 5)$  and  $(1, 7)$ . The line  $\ell_2$  has slope 2 and passes through  $(0, -1)$ . Without solving a system algebraically, argue the intersection point.

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4. (**Match to a picture in your head**) A line has slope  $-3$  and  $y$ -intercept 4. Is  $(2, -1)$

on, above, or below the line? Show the inequality check.

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5. **(Type of system)** For the system

$$y = 3x + 1, \quad 6x - 2y = 10$$

classify (one, none, infinite) and find the solution if it exists.

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6. **(Word model — bus passes)** A bus pass costs \$25 plus \$2 per ride, while a pay-as-you-go option is \$5 per ride. Let  $x$  = number of rides,  $C(x)$  = cost.

(a) Write both cost lines. (b) For what  $x$  do costs match? (c) For which  $x$  is the pass cheaper?

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7. **(Above or below?)** Consider  $y \geq -\frac{1}{2}x + 4$ . (a) Is  $(4, 1)$  a solution? (b) Is  $(0, 4)$  a solution? (c) Should the boundary be solid or dashed?

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8. **(Pick a point that works)** Give *one* point that satisfies both  $y < 2x - 1$  and  $y \geq -x + 3$ . Briefly show it works.

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9. (**From two points to a line**) The line through  $(3, -1)$  and  $(7, 5)$  bounds the half-plane  $y > mx + b$ . (a) Find  $m$  and  $b$ . (b) Is  $(0, 0)$  in the solution set?

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10. (**System meaning**) For

$$\begin{cases} y = mx + b \\ y = mx + c \quad (b \neq c) \end{cases}$$

explain the number of solutions and why *from the graph perspective*.

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11. (**Reverse engineer**) A system has the unique solution  $(2, -3)$ . Give one possible pair of line equations with *different* slopes that meet at that point.

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12. (**Feasible strip**) The region between two parallel lines  $y = 2x + 1$  and  $y = 2x - 5$  is described by a pair of inequalities. Write a correct pair using  $\leq$  or  $\geq$ .

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13. (**Which boundary is included?**) For  $y > 3x - 2$  and  $y \geq -x + 6$ , state which boundary line(s) are drawn solid and which dashed.

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14. (**Intercept thinking**) A line passes through  $(0, -2)$  and has  $x$ -intercept 4. (a) Find its

slope. (b) Determine if  $(3, 1)$  lies above or below the line.

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15. (**System via graph features**) Line  $L_1$  has slope  $-1$  and passes  $(2, 3)$ . Line  $L_2$  passes through  $(2, 3)$  and  $(6, 7)$ . (a) Write both equations. (b) What is their intersection? (No solving needed.)

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16. (**Word model — recipe**) You mix nuts: almonds at  $\$8/\text{lb}$  and peanuts at  $\$4/\text{lb}$  to get a 10-lb blend costing  $\$6/\text{lb}$ . Let  $x$  be almonds (lb),  $y$  peanuts (lb). (a) Write the two linear equations. (b) Plot-logic: explain why the solution is the intersection point.

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17. (**Quadrant reasoning**) Which inequality has the third quadrant (mostly  $x < 0$ ,  $y < 0$ ) as a solution region: (i)  $y \leq 2x + 5$  or (ii)  $y \geq 2x + 5$ ? Explain with a test point.

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18. (**Contain a given point**) Find a linear inequality whose solution set includes  $(1, 4)$  but does *not* include  $(1, 0)$ . Give one example and justify.

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19. (**Half-plane from words**) “All points at or above the line through  $(0, 2)$  with slope  $-\frac{3}{2}$ .” Write the inequality.

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20. (**All or nothing**) Suppose  $y = 2x + 7$  and  $4x - 2y = -14$ . Is the system one, none, or infinite solutions? Explain with graph logic (not just algebra).

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21. (**Closed vs open**) For  $y < \frac{3}{4}x + 2$  and  $y \geq -2x - 1$ , sketch-instructions only: state (a) which line style for each, and (b) a single point guaranteed in the intersection.

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22. (**Find any feasible integer point**) System:

$$\begin{cases} y \leq x + 1 \\ y > -2x + 4 \end{cases}$$

Give any *integer* solution  $(x, y)$  and show it works.

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23. (**Parallel trap**) Two lines have equal slopes and different  $y$ -intercepts. State the number of solutions to the system and the geometric reason.

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24. (**Design your own**) Create a system of two inequalities whose solution set is a *bounded*

(closed) quadrilateral. Write the two inequalities and name one lattice point inside.

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